

3 Huntington's disease is caused by a mutation in a gene coding for a protein that has a number of cellular functions.

(a) Identify the gene and state the name of the protein that are affected by the mutation that causes Huntington's disease.

gene

protein

[2]

(b) In many countries, a genetic screening test for Huntington's disease is available.

Adults over the age of 18 years who are considered to be at risk of developing the disease are offered the genetic screening test.

(i) Suggest which adults are considered to be at risk of developing Huntington's disease.

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(ii) Suggest **and** explain why the genetic screening test for Huntington's disease is only offered to adults who are at least 18 years old.

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..... [2]

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- (d) Some genetic diseases, such as severe combined immunodeficiency (SCID), can be treated by gene therapy.

Explain why Huntington's disease **cannot** be treated using the type of gene therapy used to treat SCID.

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